Q1------slip4----------

```
{ "brand": "Samsung",
 "model": "Galaxy A55",
 "specifications": {
 "RAM": "8 GB",
 "ROM": "128 GB",
 "processor": "Exynos 1480"
 },
"price": 39999,
 "customerRatings": [
 { "customerName": "Rahul",
 "rating": 5
 },
{ "customerName": "Priya",
 "rating": 4 },
{ "customerName": "Amit",
 "rating": 4 } ] }
```

Q2-----slip4----------

1. Create Database and Collections

```
use OnlineVehicleShopping
db.createCollection("customers")
db.createCollection("vehicles")
db.createCollection("purchases")
```

2. Insert 10 Documents in Customers Collection

```
db.customers.insertMany([
{customerId:1, name:"Amit", city:"Pune"},
{customerId:2, name:"Neha", city:"Mumbai"},
{customerId:3, name:"Rahul", city:"Nashik"},
{customerId:4, name:"Priya", city:"Nagpur"},
{customerId:5, name:"Amit", city:"Pune"},
{customerId:6, name:"Rohan", city:"Mumbai"},
{customerId:7, name:"Sneha", city:"Nashik"},
{customerId:8, name:"Vikas", city:"Pune"},
{customerId:9, name:"Pooja", city:"Kolhapur"},
{customerId:10, name:"Anjali", city:"Satara"}
])
```

3. Insert 10 Documents in Vehicles Collection

```
db.vehicles.insertMany([
{vehicleId:101, company:"Tata", model:"Nexon", type:"SUV", price:850000},
{vehicleId:102, company:"Maruti", model:"Swift", type:"Hatchback",
price:750000},
{vehicleId:103, company:"Hyundai", model:"Creta", type:"SUV",
price:1100000},
{vehicleId:104, company:"Tata", model:"Punch", type:"SUV", price:700000},
{vehicleId:105, company:"Mahindra", model:"XUV300", type:"SUV",
price:950000},
{vehicleId:106, company:"Honda", model:"City", type:"Sedan",
```

price:1200000},
{vehicleId:107, company:"Tata", model:"Altroz", type:"Hatchback",
price:800000},
{vehicleId:108, company:"Kia", model:"Seltos", type:"SUV", price:1050000},
{vehicleId:109, company:"Renault", model:"Kwid", type:"Hatchback",
price:550000},
{vehicleId:110, company:"Toyota", model:"Fortuner", type:"SUV",
price:3200000}
])
4. Insert 10 Documents in Purchases Collection
db.purchases.insertMany([
{purchaseId:1, customerId:1, vehicleId:101, rating:5},
{purchaseId:2, customerId:2, vehicleId:102, rating:4},
{purchaseId:3, customerId:3, vehicleId:103, rating:5},
{purchaseId:4, customerId:4, vehicleId:104, rating:4},
{purchaseId:5, customerId:5, vehicleId:105, rating:5},
{purchaseId:6, customerId:6, vehicleId:106, rating:3},
{purchaseId:7, customerId:7, vehicleId:107, rating:5},
{purchaseId:8, customerId:8, vehicleId:108, rating:4},
{purchaseId:9, customerId:9, vehicleId:109, rating:5},
{purchaseId:10, customerId:10, vehicleId:110, rating:4}
])
Queries
i. List all SUV vehicles
db.vehicles.find({
type:"SUV"
})
ii. List vehicles having price less than ₹10,00,000
db.vehicles.find({
price:{$lt:1000000}
})
iii. List vehicles of Tata company
db.vehicles.find({
company:"Tata"
})
iv. List customers whose name is Amit
db.customers.find({
name:"Amit"
})
v. List purchases having rating 5
db.purchases.find({
rating:5
})

Q1------slip3----------

```
{
 "personalDetails": {
 "name": "Amit Patil",
 "age": 30,
 "email": "amit@example.com",
 "phone": "9876543210"
 },
 "jobInformation": {
 "employeeId": "EMP101",
 "designation": "Software Engineer",
 "department": "IT",
 "joiningDate": "2023-06-15",
 "salary": 60000
 },
 "projects": [
 {
 "projectId": "P001",
 "projectName": "Banking Application",
 "role": "Backend Developer",
 "status": "Completed"
 },
 {
 "projectId": "P002",
 "projectName": "E-Commerce Website",
 "role": "Team Lead",
 "status": "Ongoing"
 },
 {
 "projectId": "P003",
 "projectName": "Mobile App",
 "role": "Developer",
 "status": "Planned"
 }
 ]
}
```

Q2------slip3----------

Create Database and Collections

```
 use InventoryDB
 db.createCollection("items")
 db.createCollection("warehouses")
 db.createCollection("inventory")
 2. Insert 10 Documents in Items Collection
 db.items.insertMany([
 {itemId:1, name:"Laptop", category:"Electronics", status:"A",
tags:["computer","office","portable"], height:12},
 {itemId:2, name:"Keyboard", category:"Electronics", status:"D",
tags:["computer","accessory","input"], height:5},
```

```
{itemId:3, name:"Monitor", category:"Electronics", status:"A",
tags:["display","computer"], height:15},
{itemId:4, name:"Printer", category:"Electronics", status:"D",
tags:["office","printing","computer"], height:18},
{itemId:5, name:"Chair", category:"Furniture", status:"A",
tags:["office","furniture","wood"], height:20},
{itemId:6, name:"Table", category:"Furniture", status:"A",
tags:["office","furniture"], height:25},
{itemId:7, name:"Mouse", category:"Electronics", status:"D",
tags:["computer","accessory","wireless"], height:4},
{itemId:8, name:"Scanner", category:"Electronics", status:"A",
tags:["office","scanning"], height:14},
{itemId:9, name:"Cupboard", category:"Furniture", status:"D",
tags:["storage","furniture","office"], height:30},
{itemId:10, name:"Projector", category:"Electronics", status:"A",
tags:["display","office","presentation"], height:16}
])
3. Insert 10 Documents in Warehouses Collection
db.warehouses.insertMany([
{warehouseId:101, name:"Pune Warehouse", location:"Pune"},
{warehouseId:102, name:"Mumbai Warehouse", location:"Mumbai"},
{warehouseId:103, name:"Nashik Warehouse", location:"Nashik"},
{warehouseId:104, name:"Delhi Warehouse", location:"Delhi"},
{warehouseId:105, name:"Nagpur Warehouse", location:"Nagpur"},
{warehouseId:106, name:"Surat Warehouse", location:"Surat"},
{warehouseId:107, name:"Jaipur Warehouse", location:"Jaipur"},
{warehouseId:108, name:"Chennai Warehouse", location:"Chennai"},
{warehouseId:109, name:"Bangalore Warehouse", location:"Bangalore"},
{warehouseId:110, name:"Hyderabad Warehouse", location:"Hyderabad"}
])
4. Insert 10 Documents in Inventory Collection
db.inventory.insertMany([
{itemId:1, warehouseId:101, quantity:50},
{itemId:2, warehouseId:102, quantity:25},
{itemId:3, warehouseId:103, quantity:40},
{itemId:4, warehouseId:104, quantity:60},
{itemId:5, warehouseId:105, quantity:20},
{itemId:6, warehouseId:106, quantity:35},
{itemId:7, warehouseId:107, quantity:15},
{itemId:8, warehouseId:108, quantity:45},
{itemId:9, warehouseId:109, quantity:55},
{itemId:10, warehouseId:110, quantity:28}
])
Queries
i. List all items where status is "D" and quantity is greater than 30
db.items.aggregate([
{
$lookup:{
```

```
from:"inventory",
localField:"itemId",
foreignField:"itemId",
as:"stock"
}
},
{$unwind:"$stock"},
{
$match:{
status:"D",
"stock.quantity":{$gt:30}
}
}
])
```

ii. List all items which have 3 tags

```
db.items.find({
$expr:{$eq:[{$size:"$tags"},3]}
})
```

iii. List all items having status "A" OR quantity less than 30 AND height greater than 10

```
db.items.aggregate([
{
$lookup:{
from:"inventory",
localField:"itemId",
foreignField:"itemId",
as:"stock"
}
},
{$unwind:"$stock"},
{
$match:{
$or:[
{status:"A"},
{
$and:[
{"stock.quantity":{$lt:30}},
{height:{$gt:10}}
]
}
]
}
}
])
```